

JECRC UNIVERSITY JAIPUR
Session -2025-26
Subject: Mathematics-I (DMA024A)
Course: B. Tech – I Semester
Tutorial Sheet – 13 (CO5)
UNIT-5

CO5: To acquaint the student with mathematical tools needed in evaluating multiple integrals and their usage.

PART – A (5X2=10)

A1: (CO5) Evaluate: $\int_0^2 \int_1^{2y} dy dx$.

A2: (CO5) Evaluate: $\int_0^1 \int_0^x e^{\frac{y}{x}} dy dx$

A2: (CO5) Changing the order of integration in the integral

$$I = \int_0^8 \int_{\frac{x}{4}}^2 f(x, y) dy dx \text{ leads to } I = \int_r^s \int_p^q f(x, y) dx dy. \text{ Find the value of } q.$$

A4: (CO5) Evaluate: $\int_0^3 \int_0^1 (x^2 + 3y^2) dy dx$

A5: (CO5) A triangle ABC consists of vertex points A (0, 0), B(1, 0), and C (0,1). Find the value of the integral $\iint 2xdxdy$ over the triangle ABC.

PART – B (3X7=21)

B1: (CO5) Evaluate: $\iint y dx dy$ over the region enclosed by the parabola $x^2 = y$ and the line $y = x + 2$.

B2: (CO5) Evaluate: $\int_0^1 \int_{e^x}^e \frac{1}{\log y} dy dx$

B3: (CO5) Evaluate: $\iint \frac{xy}{\sqrt{1-y^2}} dx dy$ over the first quadrant of the circle $x^2 + y^2 = 1$.

PART – C (3X11=33)

C1: (CO5) Change into polar coordinates and then solve the integral, $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dy dx$.

C2: (CO5) Evaluate $\iint_A xy dx dy$, where the region A is the positive quadrant of the circle $x^2 + y^2 = a^2$.

C2: (CO5) Change the order of integration in $I = \int_0^{2a} \int_{x^2/4a}^{3a-x} y dx dy$, and hence solve it.